

CHAPTER 8

The p-Block Elements

Group 13 Elements (Boron Family)

- Taking stability as the factor, which one of the following represents correct relationship? (2023)
 - $\text{TlI} > \text{TlI}_3$
 - $\text{TlCl}_3 > \text{TlCl}$
 - $\text{InI}_3 > \text{InI}$
 - $\text{AlCl} > \text{AlCl}_3$
- Select the element (M) whose trihalides **cannot** be hydrolysed to produce an ion of the form $[\text{M}(\text{H}_2\text{O})_6]^{3+}$ (2023-Manipur)
 - Ga
 - In
 - Al
 - B
- The correct order of atomic radii in group 13 elements is (2018)
 - $\text{B} < \text{Al} < \text{In} < \text{Ga} < \text{Tl}$
 - $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$
 - $\text{B} < \text{Ga} < \text{Al} < \text{Tl} < \text{In}$
- Which one of the following elements is unable to form MF_6^{3-} ion? (2018)
 - Ga
 - Al
 - In
 - B
- AlF_3 is soluble in HF only in presence of KF. It is due to the formation of: (2016-II)
 - $\text{K}_3[\text{AlF}_3\text{H}_3]$
 - $\text{K}_3[\text{AlF}_6]$
 - AlH_3
 - $\text{K}[\text{AlF}_3\text{H}_3]$
- The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence: (2015 Re)
 - $\text{In} < \text{Tl} < \text{Ga} < \text{Al}$
 - $\text{Ga} < \text{In} < \text{Al} < \text{Tl}$
 - $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$

Compounds of Boron, Uses of B & Al and Their Compounds

- Match List - I with List - II: (2022 Re)

	List-I (Compounds)		List-II (Molecular formula)
A.	Borax	I.	NaBO_2
B.	Kernite	II.	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$
C.	Orthoboric acid	III.	H_3BO_3
D.	Borax bead	IV.	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Choose the **correct answer** from the options given below:

- A-I, B-III, C-IV, D-II
 - A-IV, B-II, C-III, D-I
 - A-II, B-IV, C-III, D-I
 - A-III, B-I, C-IV, D-II
- $\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{heat}} \text{X} + \text{NaBO}_2$ (2022 Re)
In the above reaction the product "X" is :
 - NaB_3O_5
 - H_3BO_3
 - B_2O_3
 - $\text{Na}_2\text{B}_2\text{O}_5$

- Which of the following statement is **not correct** about diborane? (2022)
 - Both the Boron atoms are sp^2 hybridised.
 - There are two 3-centre-2-electron bonds.
 - The four terminal B-H bonds are two centre two electron bonds
 - The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
- Boric acid is an acid because its molecule: (2016 - II)
 - Accepts OH^- from water releasing proton
 - Combines with proton from water molecule
 - Contains replaceable H^+ ion
 - Gives up a proton

Group 14 Elements (Carbon Family)

- Match the following: (2020)

	Oxide		Nature
A.	CO	(i)	Basic
B.	BaO	(ii)	Neutral
C.	Al_2O_3	(iii)	Acidic
D.	Cl_2O_7	(iv)	Amphoteric

Which of the following is correct option?

- (A) (B) (C) (D)
 - (ii) (i) (iv) (iii)
 - (iii) (iv) (i) (ii)
 - (iv) (iii) (ii) (i)
 - (i) (ii) (iii) (iv)
- Which of the following oxide is amphoteric in nature? (2020-Covid)
 - SiO_2
 - GeO_2
 - CO_2
 - SnO_2
 - Which of the following species is not stable? (2019)
 - $[\text{SiF}_6]^{2-}$
 - $[\text{GeCl}_6]^{2-}$
 - $[\text{Sn}(\text{OH})_6]^{2-}$
 - $[\text{SiCl}_6]^{2-}$
 - Which of the following is incorrect statement? (2019)
 - PbF_4 is covalent in nature
 - SiCl_4 is easily hydrolysed
 - GeX_4 ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$) is more stable than GeX_2
 - SnF_4 is ionic in nature

15. It is because of inability of ns^2 electrons of the valence shell to participate in bonding that: (2017-Delhi)

- Sn^{4+} is reducing while Pb^{4+} is oxidising
- Sn^{2+} is reducing while Pb^{4+} is oxidising
- Sn^{2+} is oxidising while Pb^{4+} is reducing
- Sn^{2+} and Pb^{2+} are both oxidising and reducing

Allotropes of Carbon

16. Match List-I with List-II (2023)

List-I		List-II	
A.	Coke	I.	Carbon atoms are sp^3 hybridised.
B.	Diamond	II.	Used as a dry lubricant
C.	Fullerene	III.	Used as a reducing agent
D.	Graphite	IV.	Cage like molecules

Choose the correct answer from the options given below:

- A-III, B-IV, C-I, D-II
 - A-II, B-IV, C-I, D-III
 - A-IV, B-I, C-II, D-III
 - A-III, B-I, C-IV, D-II
17. Choose the **correct** statement: (2022)
- Both diamond and graphite are used as dry lubricants.
 - Diamond and graphite have two dimensional network
 - Diamond is covalent and graphite is ionic
 - Diamond is sp^3 hybridised and graphite is sp^2 hybridized.
18. Which of the following structure is similar to graphite? (2013)
- BN
 - B
 - B_4C
 - B_2H_6

Some Important Compounds of C & Si

19. Identify the correct statements from the following: (2020)

- $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.
 - The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
 - ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
 - CO is colorless and odourless gas.
- (1) and (3) only
 - (2) and (3) only
 - (3) and (4) only
 - (1), (2) and (3) only

20. Which of the following is not correct about carbon monoxide? (2020)

- It reduces oxygen carrying ability of blood.
- The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
- It is produced due to incomplete combustion.
- It forms carboxyhaemoglobin.

21. Which of these is not a monomer for a high molecular mass silicone polymer? (2013)

- $MeSiCl_3$
- Me_2SiCl_2
- Me_3SiCl
- $PhSiCl_3$

22. The basic structural unit of silicates is: (2013)

- SiO^-
- SiO_4^{4-}
- SiO_3^{2-}
- SiO_4^{2-}

Answer Key

- (a)
- (d)
- (c)
- (d)
- (b)
- (c)
- (b)
- (c)
- (a)
- (a)
- (a)
- (d)
- (d)
- (b)
- (d)
- (d)
- (a)
- (c)
- (b)
- (c)
- (b)